SCHOOL OF MATH

CLASS 15

Two-Step Word Problems

With the Four Operations



Read • Plan • Solve • Check

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1. Understanding the Problem

Identify what you know and what you need to find.

A museum sells **4** tickets for **$6 each.** A family pays with **$30.** **How much change should they receive?**



WHAT DO WE KNOW?	WHAT DO WE NEED TO FIND?	WHY TWO STEPS?
• 4 tickets • $6 per ticket • $30 paid	The change the family receives.	First find the total cost. Then find the change.

4 × $6 = $24 → **$30 − $24 = $6** →

Answer:
$6 in change

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Image 1 practice: Understanding the problem

Q1. A family buys 3 planetarium tickets for $7 each and pays with $25. How much change should the family receive?

Q2. A school shop has 5 packs with 4 notebooks in each pack. It sells 7 notebooks. How many notebooks remain?

Q3. Thirty-six crayons are shared equally among 6 children. One child receives 3 extra crayons. How many crayons does that child have now?

Q4. A museum opens at 9 a.m. A class buys 5 tickets at $4 each and pays with $30. How much change does the class receive?

Q5. A reading corner has 32 books. A teacher adds 14 books, and children borrow 9. How many books are there now?

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2. Planning Two Steps

Decide what to find first. Use that answer in the next step.

A gardener has **3** trays with **8 flowers** in each tray. She plants **10** flowers. **How many flowers are left?**



1 STEP 1: FIND THE TOTAL

Multiply the number of trays by the flowers in each tray.

3 × 8 = 24 flowers

2 STEP 2: FIND WHAT IS LEFT

Subtract the flowers planted from the total.

24 – 10 = 14 flowers

Answer:

14 flowers are left.

The answer to Step 1 becomes information for Step 2.

Image 2 practice: Planning two steps

Q6. A gardener has 4 trays with 7 seedlings in each tray. She plants 9 seedlings. How many seedlings are left? State what you will find first.

Q7. A craft club has 18 red beads and 12 blue beads. It shares all the beads equally among 5 children. How many beads does each child receive?

Q8. A baker makes 48 rolls and sets aside 8 for a display. The rest go equally into 5 bags. How many rolls go into each bag?

Q9. Thirty-two bookmarks are shared equally among 4 tables. A teacher adds 5 more bookmarks to one table. How many bookmarks are on that table?

Q10. A school orders 6 boxes with 5 pencils in each box. It also receives 8 loose pencils. How many pencils arrive altogether?



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3. Choosing the Operations

Think about what is happening in the story.

+ ADD

Combine quantities or find a total.



12 visitors + **5** visitors = **17** visitors

− SUBTRACT

Take away or find the difference.



17 visitors − **4** visitors = **13** visitors

× MULTIPLY

Find the total in equal groups.



3 rows × **6** seats per row = **18** seats

÷ DIVIDE

Share equally or find equal groups.



18 students ÷ **3** equal groups = **6** students per group

PUT TWO OPERATIONS TOGETHER

A gallery has **4** rooms with **5** paintings in each. It adds **3** more paintings. How many paintings are there now?



Step 1: Multiply.
4 × 5 = 20 paintings.

→

Step 2: Add.
20 + 3 = 23 paintings.

Answer:
23 paintings.

Choose operations from the situation, not just from key words.

Image 3 practice: Choosing the operations

Q11. A club has 16 members. Seven new members join and 5 members leave. How many members are there now? Name both operations.

Q12. Five tables each need 6 chairs. A teacher brings 4 extra chairs for guests. How many chairs are needed in all? Name both operations.

Q13. Forty-two cards are shared equally among 7 students. One student gives away 2 of her cards. How many cards does she keep? Name both operations.

Q14. A shop has 27 orange balloons and 18 green balloons. It puts all the balloons into bunches of 5. How many bunches can it make? Name both operations.

Q15. Liam has 4 bags with 6 marbles in each bag. Ava has 31 marbles. How many more marbles does Ava have than Liam?



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4. Addition and Subtraction

Combine quantities, then find what remains.

A library has **28** books on a cart.
A librarian adds **15** more books.
Students borrow **19** books.
How many books remain on the cart?



STEP 1: ADD THE BOOKS

Find the total after the librarian adds more.

$$28 + 15 = 43 \text{ books}$$



STEP 2: SUBTRACT THE BOOKS BORROWED

Use the total from Step 1.

$$43 - 19 = 24 \text{ books}$$



Answer:

24 books remain on the cart.

CHECK YOUR ANSWER

Undo the steps in reverse order.

$24 + 19 = 43$

$43 - 15 = 28$

Image 4 practice: Addition and subtraction

Q16. A library cart has 37 books. A librarian adds 26 books, and students borrow 18. How many books remain on the cart?

Q17. A school bus has 45 passengers. Seventeen get off, then 9 get on. How many passengers are now on the bus?

Q18. A shop has 68 postcards, sells 24, and receives 15 more. How many postcards does it have now?

Q19. A reading challenge begins with 86 points. A team earns 27 points and then spends 35 points on a reward. How many points does the team have left?

Q20. A class collects 125 cans on Monday and 48 on Tuesday. It sends 67 cans for recycling. How many cans are waiting to be sent?



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5. Multiplication, Then Addition or Subtraction

Find the total in equal groups. Then adjust the total.

MULTIPLY, THEN ADD

A concert hall has **6** rows with **8** seats in each row. Staff add **5** extra chairs. How many seats are available?



Step 1: $6 \times 8 = 48$ seats

Step 2: $48 + 5 = 53$ seats

Answer:

53 seats are available.

MULTIPLY, THEN SUBTRACT

A music school has **4** boxes with **9** tambourines in each box. Students take **7** tambourines. How many tambourines remain?



Step 1: $4 \times 9 = 36$ tambourines

Step 2: $36 - 7 = 29$ tambourines

Answer:

29 tambourines remain.

KEY IDEA: TWO-STEP WORD PROBLEMS

Equal groups	Extras	Taken away
→ Multiply	→ Add	→ Subtract

Image 5 practice: Multiply, then add or subtract

Q21. A music room has 7 rows with 6 chairs in each row. Staff add 9 extra chairs altogether. How many chairs are there now?

Q22. A coach has 5 boxes with 8 tennis balls in each box. Students take 13 balls to practice. How many balls remain?

Q23. A shop receives 8 packs with 7 stickers in each pack and 6 loose stickers. How many stickers arrive?

Q24. A family buys 4 art kits costing $9 each and pays with $50. How much change should it receive?

Q25. A garden has 9 rows with 8 plants in each row. Gardeners move 17 plants to another garden. How many plants remain?



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6. Division, Then Addition or Subtraction

Share equally. Then adjust one share.

+ DIVIDE, THEN ADD

An art teacher shares **24** paintbrushes equally among **4** tables. She puts **2** extra paintbrushes on one table.
How many paintbrushes are on that table now?



Step 1: $24 \div 4 = 6$ paintbrushes per table

Step 2: $6 + 2 = 8$ paintbrushes

Answer:
8 paintbrushes are on that table.

− DIVIDE, THEN SUBTRACT

A pottery teacher shares **30** clay tools equally among **5** students. One student returns **2** tools.
How many tools does that student keep?



Step 1: $30 \div 5 = 6$ tools per student

Step 2: $6 - 2 = 4$ tools

Answer:
That student keeps 4 tools.

KEY IDEA: TWO-STEP WORD PROBLEMS

Equal sharing	Added to one share	Removed from one share
→ **Divide**	→ **Add**	→ **Subtract**

Image 6 practice: Divide, then adjust one share

Q26. Forty-eight paintbrushes are shared equally among 6 tables. One table receives 3 extra brushes. How many brushes are on that table now?

Q27. Forty-five clay tools are shared equally among 5 students. One student returns 4 tools. How many tools does that student keep?

Q28. Fifty-six beads are shared equally among 7 children. One child finds 6 more beads. How many beads does that child have?

Q29. Sixty-three sheets of paper are shared equally among 9 groups. One group uses 2 sheets. How many unused sheets does that group have?

Q30. Eighty-one counters are shared equally among 9 students. One student receives 7 bonus counters. How many counters does that student have now?



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6. Division, Then Addition or Subtraction

Share equally. Then adjust one share.

+ DIVIDE, THEN ADD

An art teacher shares **24** paintbrushes equally among **4** tables. She puts **2** extra paintbrushes on one table.
How many paintbrushes are on that table now?



Step 1: $24 \div 4 = 6$ paintbrushes per table

Step 2: $6 + 2 = 8$ paintbrushes

Answer:
8 paintbrushes are on that table.

− DIVIDE, THEN SUBTRACT

A pottery teacher shares **30** clay tools equally among **5** students. One student returns **2** tools.
How many tools does that student keep?



Step 1: $30 \div 5 = 6$ tools per student

Step 2: $6 - 2 = 4$ tools

Answer:
That student keeps 4 tools.

KEY IDEA: TWO-STEP WORD PROBLEMS

Equal sharing	Added to one share	Removed from one share
→ **Divide**	→ **Add**	→ **Subtract**

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Image 7 practice: More practice with equal shares

Q31. Seventy-two pencils are shared equally among 8 tables. One table gets 4 extra pencils. How many pencils does it have now?

Q32. Sixty-four cards are shared equally among 8 players. One player gives away 3 cards. How many cards does that player keep?

Q33. Ninety-six stickers are shared equally among 8 children. One child earns 5 extra stickers. How many stickers does that child have?

Q34. One hundred eight tiles are shared equally among 9 groups. One group sets aside 4 of its tiles. How many of its tiles are not set aside?

Q35. Eighty-four game tokens are shared equally among 7 players. One player spends 5 tokens. How many tokens does that player have left?



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8. Writing and Checking the Answer

Show both steps, label units, and check your result.

A museum shop has 5 boxes with 6 postcards in each box. It sells 8 postcards. How many postcards remain?



1 SHOW BOTH CALCULATIONS

Step 1: $5 \times 6 = 30$ postcards

Step 2: $30 - 8 = 22$ postcards



2 WRITE A COMPLETE ANSWER

22 postcards remain in the museum shop.



3 CHECK WITH INVERSE OPERATIONS

Undo the steps in reverse order.

$22 + 8 = 30$ postcards

$30 \div 5 = 6$ postcards per box



4 DOES IT MAKE SENSE?

22 is less than **30** because some postcards were sold.



CHECK YOUR WORK

- ☑ Both steps shown
- ☑ Units included
- ☑ Answer checked

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Image 8 practice: Writing and checking the answer

Q36. A gift shop has 6 boxes with 7 badges in each box. It sells 10 badges. How many badges remain? Show both steps and check.

Q37. A team receives 54 ribbons to share equally among 6 children. One child earns 4 more ribbons. How many ribbons does that child have? Check your answer.

Q38. A library has 76 magazines, receives 18 more, and lends out 29. How many magazines remain? Include a unit and a reverse check.

Q39. A shop puts 25 red pencils and 35 blue pencils equally into 6 bags. How many pencils go in each bag? Show and check both steps.

Q40. A family buys 7 tickets costing $8 each and pays with $70. A student says the change is $56. Find the correct change and explain what $56 represents.

Homework review: R1–R5

R1. A family buys 5 aquarium tickets at $7 each and pays with $40. What must you find before the change, and how much change should the family receive?

R2. A gardener has 6 trays with 4 seedlings in each tray and plants 11 seedlings. State a two-step plan and find how many are left.

R3. A club collects 21 red counters and 15 yellow counters. It shares them equally among 6 children. Name the operations and find each child’s share.

R4. A librarian starts with 54 books, adds 17 books, and lends out 26. How many books remain?

R5. A music room has 8 rows with 5 chairs in each row. Staff bring in 7 extra chairs altogether. How many chairs are available?

Homework review: R6–R8

R6. Forty-nine pencils are shared equally among 7 students. One student receives 5 extra pencils. How many pencils does that student have?

R7. Seventy-seven tokens are shared equally among 7 players. One player spends 4 tokens. How many tokens does that player keep? Check in reverse.

R8. A shop has 9 packs with 6 bookmarks in each pack and sells 16 bookmarks. How many bookmarks remain? Write both steps and a reverse check.

Homework word problems: W1–W5

W1. A class has 23 markers. A teacher brings 12 more, and 8 markers run out of ink. How many markers still work?

W2. A bakery fills 4 trays with 6 muffins each. It sells 9 muffins. How many muffins remain?

W3. An art club fills 5 tables with 7 children at each table. Three more children arrive. How many children are there altogether?

W4. Thirty-five stickers are shared equally among 5 friends. One friend gets 4 extra stickers. How many stickers does that friend have?

W5. Fifty-four crackers are shared equally among 6 children. One child eats 3 of her crackers. How many crackers does she have left?

Homework word problems: W6–W10

W6. A family buys 6 sketchbooks for $7 each and pays with $50. How much change does it receive?

W7. A class collects 24 cans in the morning and 32 in the afternoon. It packs all the cans into boxes of 8. How many boxes does it fill?

W8. A bakery has 72 rolls. It saves 16 for a school lunch and puts the rest equally into 7 bags. How many rolls go in each bag?

W9. Maya has 7 packs with 8 cards in each pack. Noah has 68 cards. How many more cards does Noah have than Maya?

W10. Three class projects earn 18 points each. The class needs 80 points for a reward. How many more points does it need?

Homework word problems: W11–W12

W11. A school buys 8 boxes with 12 pencils in each box. It shares all the pencils equally among 6 classrooms. How many pencils does each classroom receive?

W12. A teacher gives the same number of beads to each of 8 children. Each child gets 9 beads, and 14 beads remain in the box. How many beads did the teacher have before sharing?